

Farhan Akhtar Makandar

Agentic AI Engineer

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PROFESSIONAL SUMMARY

Agentic AI Engineer specializing in designing and deploying autonomous AI agent systems, multi-agent workflows, and RAG pipelines. NVIDIA-certified in Building RAG Agents with LLMs. Proven expertise in LangChain, vector databases, prompt engineering, and LLM reasoning chains. Experienced in building intelligent systems that combine retrieval, reasoning, and generation for complex document understanding and decision-making tasks. Strong foundation in transformer architectures and production ML system design.

TECHNICAL SKILLS

- Agentic AI: Multi-Agent Systems, Autonomous Agents, Tool Use, Function Calling, Planning
- RAG: Retrieval-Augmented Generation, Vector Databases, Semantic Search, Chunking Strategies
- LLM: Prompt Engineering, Chain-of-Thought, ReAct, LLM Integration, Fine-Tuning
- Frameworks: LangChain, LLM Orchestrators, Agent Frameworks, Workflow Engines
- ML/DL: PyTorch, TensorFlow, Transformers, CNN, RNN, LSTM, Attention Mechanisms
- Data: Pandas, NumPy, Feature Engineering, Embeddings, Similarity Search
- Cloud: AWS, Oracle Cloud, MongoDB, Git/GitHub

PROFESSIONAL EXPERIENCE

Research Intern - AI/ML - NIT Calicut

Jun 2024 - Dec 2024

- Developed and evaluated transformer-based models for complex predictive tasks using PyTorch, achieving rigorous performance benchmarks
- Collaborated on cutting-edge AI research focused on neural network optimization and deep learning applications for engineering problems
- Conducted comprehensive data analysis, feature engineering, and model validation using scientific computing libraries
- Assisted in documenting research findings and preparing technical reports for publication

PROJECTS

Agentic AI System - Multi-Step RAG Pipeline

Architected end-to-end agentic RAG system combining document retrieval via vector database, multi-step LLM reasoning, and context-aware response generation. Implemented agent planning with tool use for complex queries requiring multi-document synthesis.

Techn

Seismic Response Prediction - TFT

Built Temporal Fusion Transformer with attention-based architecture for time-series prediction. Applied multi-head attention and variable selection networks for interpretable predictions.

Techn

SAINT Model - Dual Attention Architecture

Implemented Self-Attention and Intersample Attention mechanism, demonstrating expertise in attention architectures applicable to agentic reasoning and information retrieval systems.

Techn

EDUCATION

B.E. in Artificial Intelligence & Machine Learning

Yenepoya Institute of Technology, Moodbidri (VTU) | Expected 2026

CERTIFICATIONS

- IBM Python for Data Science (Credly Verified)
- Machine Learning with Python - IBM Cognitive Class
- Deep Learning Fundamentals - IBM Cognitive Class
- Accelerating Deep Learning with GPUs - IBM Cognitive Class
- Supervised Machine Learning - Coursera
- Building RAG Agents with LLMs - NVIDIA
- AWS Generative AI Essentials - AWS
- AWS Prompt Engineering - AWS
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Accenture Data Analytics Simulation - Forage
- MongoDB Atlas - MongoDB University
- HTML Training - IIT Bombay